MIL-P-37797A
28 September 1978
SUPERSEDING
MIL-P-37797(DLA-DM)
30 November 1977

MILITARY SPECIFICATION

PLUGGER SET, PLASTIC FILLING, DENTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a dental plugger set and replacement tips suitable for placement of composite resin restorative material.

1.2 Classification. The plugger set and replacement parts shall be of the following types, as specified (see 6.1):

Type I. - Plugger set, complete (with tip assortment).
Type II. - Tip assortment (replacement tips).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods Of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24392 - Plugger Set, Plastic Filling, Dental.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 - Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E 92 - Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General requirements. The pluggers shall be in accordance with the applicable requirements of GG-I-526 with the options and additional requirements specified herein

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3.2 Material. The pluggers and tips shall be fabricated of the materials specified in table I.

TABLE I. Materials.

Classification	Component	Material	Specification
Type I	Pluggers	Stainless Steel	
	Handles	Class 4, type 440A	GG-I-526
	Operative ends (tubing)	Class 3, type 303	GG-I-526
Types I and II	Tip assortment (plastic tips)	Acetyl homopolymer	-

3.3 Style, design, and dimensions. Style, design, and dimensions of the type I plugger set and type II tip assortment shall be in conformance with Defense Personnel Support Center (DPSC) Drawing No. 24392.

3.3.1 Type I plugger set, complete (with tip assortment). The plugger set shall be suitable for placing and condensing composite resin materials in natural dentition. The plugger shall be suitable for use with type II tips.

3.3.1.1 Plugger construction. The plugger shall be a double-ended instrument of three-piece construction. The handle shall be solid, octagonal in shape, with the operative ends consisting of corrosion-resistant steel tubing, soldered with silver solder to the ends of the shanks of the handle. The tubing shall be securely soldered to the shank ends of the handle. The operative ends (tubing) of each instrument shall face in opposite directions. The operative ends shall accept the interchangeable plastic tips.

3.3.2 Type II, tip assortment (replacement tips). The tip assortment shall be suitable for use with pluggers of type I plugger set. The tips shall exhibit non-sticking qualities and shall not discolor or contaminate composite restorative materials.

3.3.2.1 Tip construction. Tips shall be so designed as to be removable and interchangeable with the operative ends (steel tubing) of the pluggers. The tips shall provide a press fit with respect to the tubing of the pluggers.

3.4 Hardness. The handles shall have a Rockwell hardness of C-52 to C-56. The operative end tubing shall be work-hardened to provide a hardness suitable for the purpose intended.

3.5 Corrosion resistance. Pluggers shall meet the requirements of GG-I-526 for resistance to copper sulfate and resistance to boil for the applicable material.

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3.6 Finish.

3.6.1 Pluggers. Handles, including shanks and operative ends, shall have a mirror or bright polished finish. The inside of the operative ends (steel tubing) shall have a smooth finish. Handles shall have 40 + 10 fillet or notch-type serrations per inch on seven faces, and shall have one smooth face for identification marking. The depth of the serrations shall be 0.001 to 0.004 inch.

3.6.2 Tips. The blade portion of the plastic tips shall be smooth and free of parting lines and fins. The shanks of the tips shall have a smooth finish.

3.7 Quantity. Each unit of issue shall consist of the components listed in table II.

TABLE II. Components and quantity in unit of issue.

Classification	Component	Quantity
Type I	Plugger	2 each
	Tips	
	Spade (small)	4 each
	Spade (large)	4 each
	Cylindrical	4 each
Type II	Tips	
	Spade (small)	20 each
	Spade (large)	20 each
	Cylindrical	20 each

3.8 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with a proper design, configuration, fit, and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

3.9 Identification marking. Pluggers shall be permanently and legibly marked in accordance with GG-I-526 and as specified herein. Pluggers shall be marked with the identification marking "F1" on one end of the handle, on the smooth face. The word "stainless" shall also be marked on instruments of foreign manufacture. As an alternate to the word "stainless", the use of a registered trademark identifying the steel as stainless shall be acceptable, provided prior approval is received from the cognizant contracting officer.

3.10 Workmanship. Workmanship shall be first class throughout. The plugger sets and tip assortments, as applicable, shall be free from defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary, to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel and plastic materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. The plugger sets or tip assortments, as applicable, shall be examined for defects including, but not limited to, those listed in table III. Examination of defects not readily perceptible to the naked eye shall be under four power (4X) magnification.

TABLE III. Classification of defects.

Categories	Defects
Major A:	
101	Type not as specified.
102	Style, design, and construction not as specified for applicable type.
103	Tips fail to fit properly in operative ends of plugger.
104	Tubing not securely soldered to shanks of handle.
Major B:	
151	Dimensions of operative ends of plugger or shank ends of tips not as specified.

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TABLE III. Classification of defects. - (Continued)

Categories	Defects
Minor A:	
201	Finish not as specified.
202	Identification marking on handle missing, inadequate, inaccurate, or illegible.
Minor B:	
251	Any dimension (except 151) not as specified.
252	Quantities of pluggers or tips furnished not as specified for type I or II.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table IV. The unit of product for examination purposes shall be one type I plugger set or one type II tip assortment, as applicable.

TABLE IV. Examination criteria.

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-1	2.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Testing. Pluggers shall be tested as specified in table V and 4.3. Inspection levels and acceptable quality levles (AQL's) shall be as listed in table V. The unit of product for test purposes shall be one plugger.

TABLE V. Tests.

Component	Characteristic	Require-ment	Test procedure	Inspection level	AQL
Plugger handle and tubing	Chemical composition	3.2	GG-I-526	*	**
Handle	Copper sulfate	3.4,3.5 and	4.3.2 and		
	Hardness	GG-I-526	GG-I-526	S-2	1.0
Plugger	Boil test	3.5 and GG-I-526	GG-I-526	S-2	1.0

* A sufficient quantity of each lot of metal used in the construction of the pluggers shall be selected for composition analysis in lieu of sample pluggers, provided lot identity has been maintained and can be demonstrated to the Government.

** Acceptance number shall be zero for all sample sizes.

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4.2.2.1 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Chemical analysis. In the absence of certificates of quality for the steel, chemical analysis shall be conducted in accordance with method 111 or method 112 of FED-STD-151.

4.3.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.4 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Each plugger set and replacement tip assortment shall be cleaned in accordance with MIL-P-116, method C-1.

5.1.1.2 Drying. Plugger sets and tip assortments (replacement tips) shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging.

5.1.1.4.1 Complete set. Four each of the small spade, large spade, and cylindrical shaped tips shall be packaged in a sealed plastic bag or vial. The packaged tips along with two instruments shall be packaged in accordance with MIL-P-116, method IC-1.

5.1.1.4.2 Replacement parts (tip assortment). The tip assortment consisting of twenty small spade tips, twenty large spade tips, and twenty cylindrical shaped tips shall be packaged in a plastic or paperboard box of an appropriate size and design. The tips shall be separated as to type within the box by partitions. As an alternate, twenty tips of one type only shall be packaged in a clear, sealed plastic bag. Three bags containing small spade tips, large spade tips, and cylindrical shaped tips shall be packaged in a sealed plastic bag.

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5.1.1.5 Intermediate package. Twelve unit packages containing complete sets or replacement parts (tip assortments) shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. Each complete set, consisting of four each of the small spade, large spade, and cylindrical shaped tips, along with two instruments, or one tip assortment, consisting of twenty each of the small spade, large spade, and cylindrical shaped tips, shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Twelve intermediate packages containing complete sets or replacement parts (tip assortments) shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve intermediate packages containing complete sets or replacement parts (tip assortments) shall be packaged in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged sets or replacement parts shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit packages, intermediate packages, and shipping containers. Each unit package, intermediate package, and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation
6520-01-035-5572	TIP ASSORTMENT, PLASTIC FILLING PLUGGER, Plastic, 60s.	Type II
6520-01-036-4523	PLUGGER SET, PLASTIC FILLING, DENTAL, Corrosion-Resisting Steel.	Type I

6.3 Approval of trade name, if applicable (see 3.9).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-37797A PLUGGER SET, PLASTIC FILLING, DENTAL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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